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# Gen AI Architect Program

Assignment 1 - PyAutoGUI Automation | learn. build. operate

## 1. Scenario

You are working as an operations executive. Every morning you need to prepare a daily status report. Your task is to build a **PyAutoGUI bot** that prepares this report automatically by controlling the browser and a spreadsheet application, just like a person would - moving the mouse, clicking, typing, and copying data on screen. This is a **medium-level** automation assignment.

*Difficulty: Medium · Tools: PyAutoGUI, Chrome, Microsoft Excel (or Numbers on Mac)*

## 2. What your bot must do

The bot should perform the following sequence on its own:

- Open Chrome and go to any public website (for example: a weather, news, or stock-price site).
- Copy the important piece of information from the page (the temperature, the top headline, or a stock value).
- Open Microsoft Excel (or Numbers on Mac).
- Create a new row containing three things: today's date & time, the fetched data, and your own short comment (for example, “Good for outdoor activities”).
- Save the Excel file with today's date in the filename, e.g. `daily_report_2025-06-17.xlsx`.
- Take a screenshot of the final Excel sheet and save it.

> **Think it through**
> PyAutoGUI controls the real mouse and keyboard, so timing and screen position matter. Think about how your bot waits for apps to open, how it makes sure the right window is focused, and what happens if something loads slower than expected. Plan the on-screen steps before you write them.

## 3. Requirements

- Use **PyAutoGUI** as the core automation library. You may add helper libraries you decide you need.
- The date and time in the row must be generated automatically at run time - not typed by hand.
- The saved filename must include the current date in the `YYYY-MM-DD` format.
- The bot must save both the Excel file and a screenshot of the final sheet.
- Keep all your code in a single file named `daily_report_bot.py`.

## 4. Step-by-step setup

Follow these steps in order. Commands are shown for both macOS/Linux and Windows where they differ.

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### Step 1 - Create a project folder

```
mkdir daily-report-bot
cd daily-report-bot
```

### Step 2 - Create a virtual environment (venv)

A virtual environment keeps this project isolated from the rest of your system.

```
python -m venv venv
# (on some systems use 'python3' instead of 'python')
```

### Step 3 - Activate the virtual environment

```
# macOS / Linux:
source venv/bin/activate

# Windows (PowerShell):
venv\Scripts\Activate.ps1

# Windows (Command Prompt):
venv\Scripts\activate.bat
```

When activated, you will see `(venv)` at the start of your terminal prompt.

### Step 4 - Install the required libraries

Install PyAutoGUI (and any helper libraries you choose to use) inside the activated venv.

```
pip install pyautogui
# add any other libraries your bot needs, e.g.:
# pip install pyperclip openpyxl
```

### Step 5 - Run your program

```
python file_name.py
```

Do not move the mouse or type while the bot is running - it is controlling them for you.

### Step 6 - Deactivate when finished

```
deactivate
```

## 5. Expected result

After a successful run you should have an Excel file named with today's date that contains a new row with the date & time, the data your bot fetched, and your comment - plus a saved screenshot of that final sheet.

## 6. What to submit

- Your `daily_report_bot.py` file.
- A short screen recording (30–60 seconds) showing the bot working.
- A sample output Excel file and the saved screenshot.
- Upload the code to GitHub and send us the repository link.

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Reminder: build the automation logic yourself - the steps above are your setup roadmap, not the answer. Good luck!